



HOW DEPRIVATION RELATES TO FOOD HYGIENE STANDARDS

CMP419: Advanced Topics in Computing



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Introduction

For context this project was developed to allow users to explore whether there is a relationship between food hygiene ratings and area deprivation. We use two separate datasets, “Food Hygiene Rating Scheme” and the “Index of Multiple Deprivation 2025”. These were both cleaned, joined and exported through Tableau Prep Builder, and visualisation was through Tableau Cloud.

The main focus of the project was to create a fully automated workflow that can take raw data, process it and clean, output the clean data into a new dataset, then visualise the data with informative graphs using Tableau Cloud.

The aim is to provide a simple tool that can explore the patterns between deprivation and lower hygiene ratings for each local authority area.

Dataset Choice

The first dataset I used was the Food Hygiene Rating Scheme dataset, and this contains the inspection information on food businesses, including the business name, business type, local authority, address, postcode, latitude, longitude, hygiene rating, hygiene score, and structural score. I chose this dataset because it is large, very detailed and is linked to a real issue with food safety.

The second dataset I used was the Index of Multiple Deprivation 2025 local authority dataset, which contains deprivation measurements for local authorities in England. The main fields I used were the IMD average score and IMD average rank, which I renamed to Deprivation Score and Deprivation Rank in the final dataset output. I decided to choose this dataset because it allowed me to compare food hygiene ratings against social and economic deprivation in different locations.

I feel as if both these datasets work well together as they measure two different but potentially related issues, where the FHRS dataset shows the individual food business hygiene inspection scores, while the IMD dataset shows how deprived an area is overall. Joining both of these datasets together allowed exploring whether food hygiene standards vary across different areas with lower or higher deprivation levels.

The FHRS dataset did require more cleaning than the deprivation dataset as there was one issue in the rating field that contained string values such as a business awaiting inspection. I decided these records could not really be used in the overall score for the hygiene and filtered them out. There were also some missing values in location fields, especially the latitude and longitude which affected some of the map visualisation because business could not be plotted on the map without their co-ordinates.

Another issue was that the two datasets did not always name local authorities the exact same way. An example of this was that one dataset might include punctuation or using “City” or “City of” in their local authority because the join depended on the local authority names matching correctly, I had to clean these fields before joining.

Figure 1 Both datasets loaded into Tableau Prep, including FHRS and IMD 2025 data

The screenshot displays the Tableau Prep interface. On the left, the 'Connections' pane lists 'FHRS_All_en-GB.csv' and 'File_10_-_IoD2025_L...' as text files. Below, the 'Tables' pane shows a list of categories including Barriers, Crime, Education, Employment, Health, IDACI, IDAOPI, and IMD. The main workspace shows a data flow: 'FHRS_All_en-GB' is connected to 'Clean 1', and 'IMD' is connected to 'Clean 2'. Both 'Clean 1' and 'Clean 2' feed into 'Join 1'. 'Join 1' then feeds into 'Clean 3', which finally leads to the 'Output' stage. The bottom pane shows the 'Settings' for the 'FHRS_All_en-GB' connection, including 'Connection Details' (Text file, File path), 'Header Options' (Set header), and 'Header and Data Start Row' (Header row 1, Data start row 2). On the right, the 'Data Preview' pane shows a table with columns: Type, Field Name, Changes, and Preview. The table lists fields like Source Row Number, AddressLine1, AddressLine2, AddressLine3, AddressLine4, BusinessTypeID, FHRSID, BusinessName, and BusinessType.

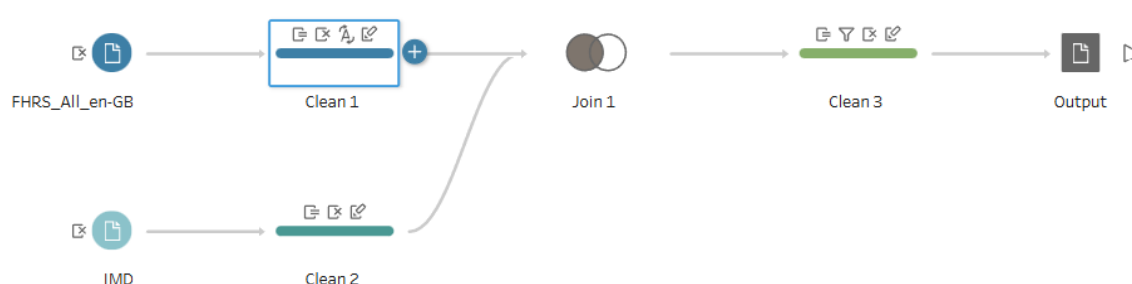
Type	Field Name	Changes	Preview
#	Source Row Number		65,537; 65,538; 65,539
Abc	AddressLine1		C Of E Primary School, Orc...
Abc	AddressLine2		School Lane, Grange Drive...
Abc	AddressLine3		Newbold, Castle Doningto...
Abc	AddressLine4		Coalville, Derby
#	BusinessTypeID		7,845
Abc	FHRSID		1,932,390, 1,683,498, 1,6...
Abc	BusinessName		Relish School Food - Newb...
Abc	BusinessType		School/college/university
Abc	BusinessAddress		null, 0, 5

Processing and Sanitisation

The data processing was all completed through Tableau Prep Builder, and I chose this tool because it allowed me to easily create a repeatable and automated flow for loading, cleaning, joining and exporting the data. This was important because data can change and be updated, so once the flow was built it could re-run and all the same cleaning steps could be applied again with new data.

The entire workflow starts with one CSV file and an Excel Spreadsheet which was the FHRS CSV file and the IMD 2025 Excel spreadsheet. Each dataset has its own cleaning stage before the join which made it easier because each dataset had different problems that needed to be fixed.

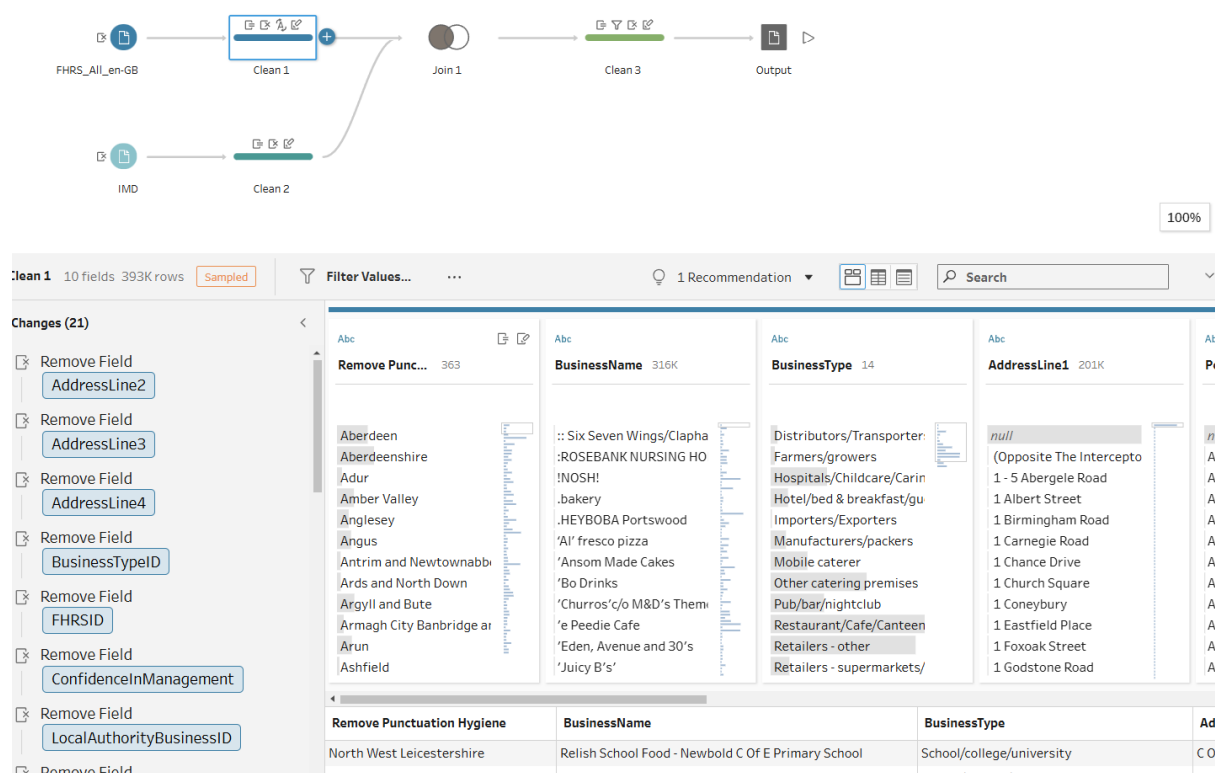
Figure 2 Process of each stage



For the FHRS dataset I decided to remove fields that were not needed for the final output. These included the fields such as the extra address lines, business IDs, local authority codes, ratings keys, right to reply, scheme type and management confidence. I removed these as they basically duplicated other data in a different way, and they did not help answer the main research question. Keeping the relevant fields reduced the noise in the final dataset and made visualisation easier to configure.

As said with the issues in the introduction the hygiene rating field was changed into a numerical format, because Tableau required numeric values to calculate averages. Some businesses had not undergone their rating, or were waiting for their rating, this was not useful for the visualisations as it provided no usable information. This kept the field consistent with only numerical values that I could use as useful data.

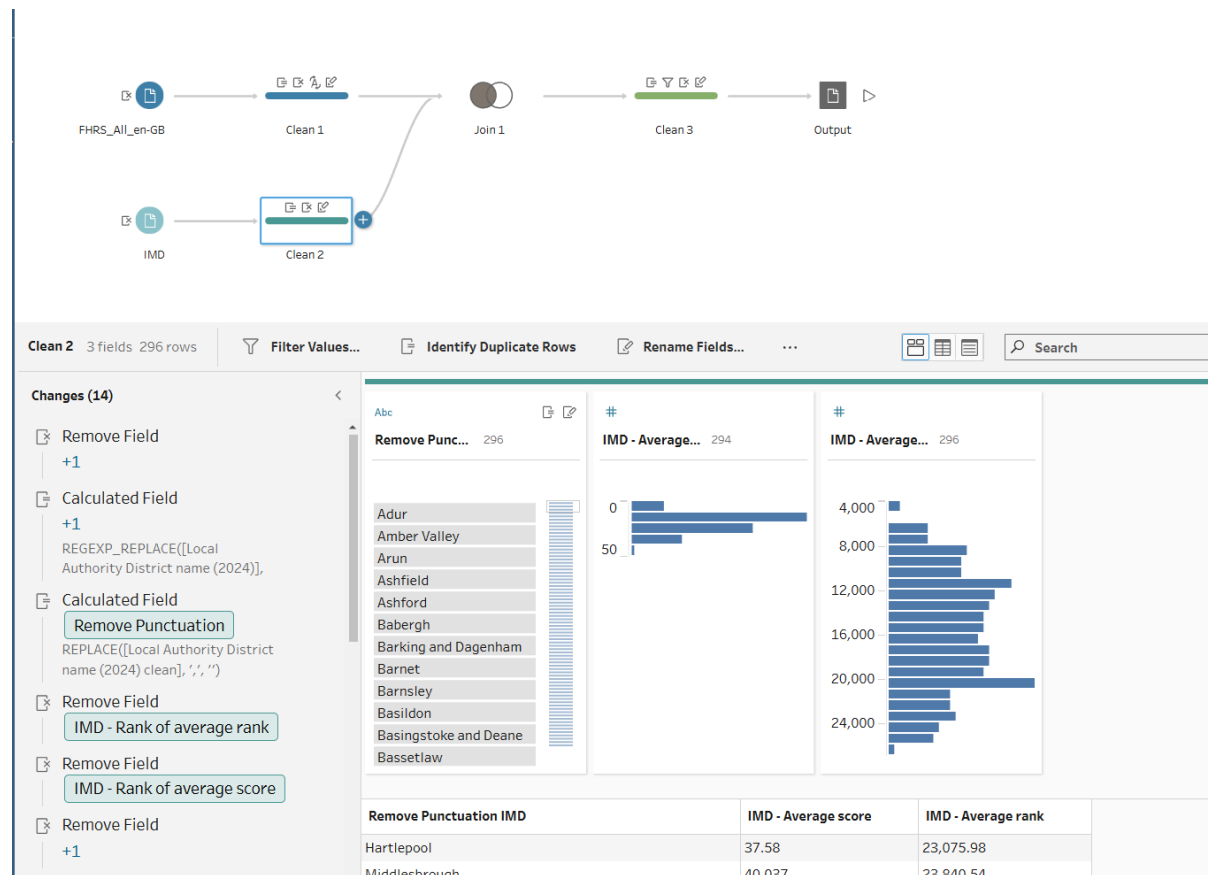
Figure 3 FHRS Cleaning stage (Clean 1)



The IMD dataset was also cleaned to remove fields that were not required, such as extra ranking measurements and local concentration fields. I kept the local authority’s name, average deprivation score and the average deprivation rank because these were the fields needed for comparison with the food hygiene data.

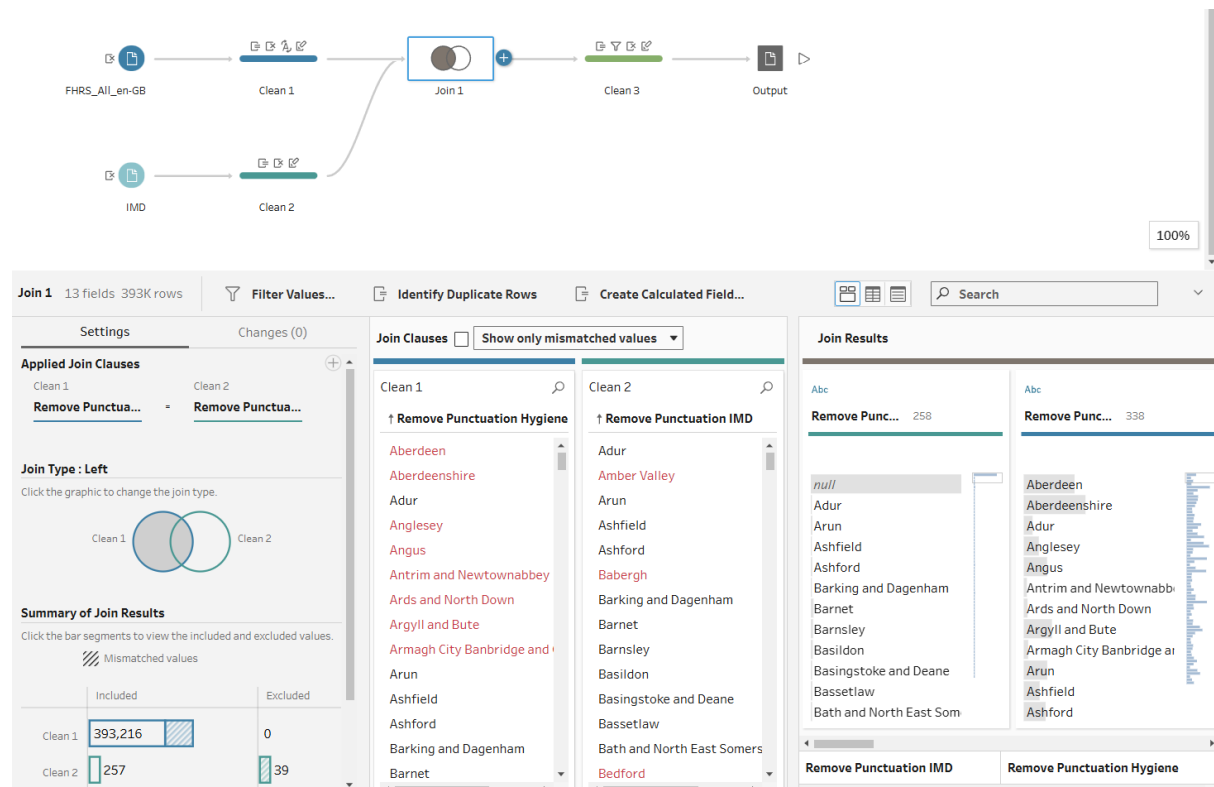
A key part of the cleaning stages for each dataset was cleaning the local authority names. I created calculated fields in Tableau Prep and used the functions “REGEXP_REPLACE” and “REPLACE”. These were used to remove wording such as “City of” and punctuation like commas. This was all done to help standardise the local authority names so the FHRS and IMD dataset could be joined more accurately.

Figure 4 IMD Cleaning stage (Clean 2)



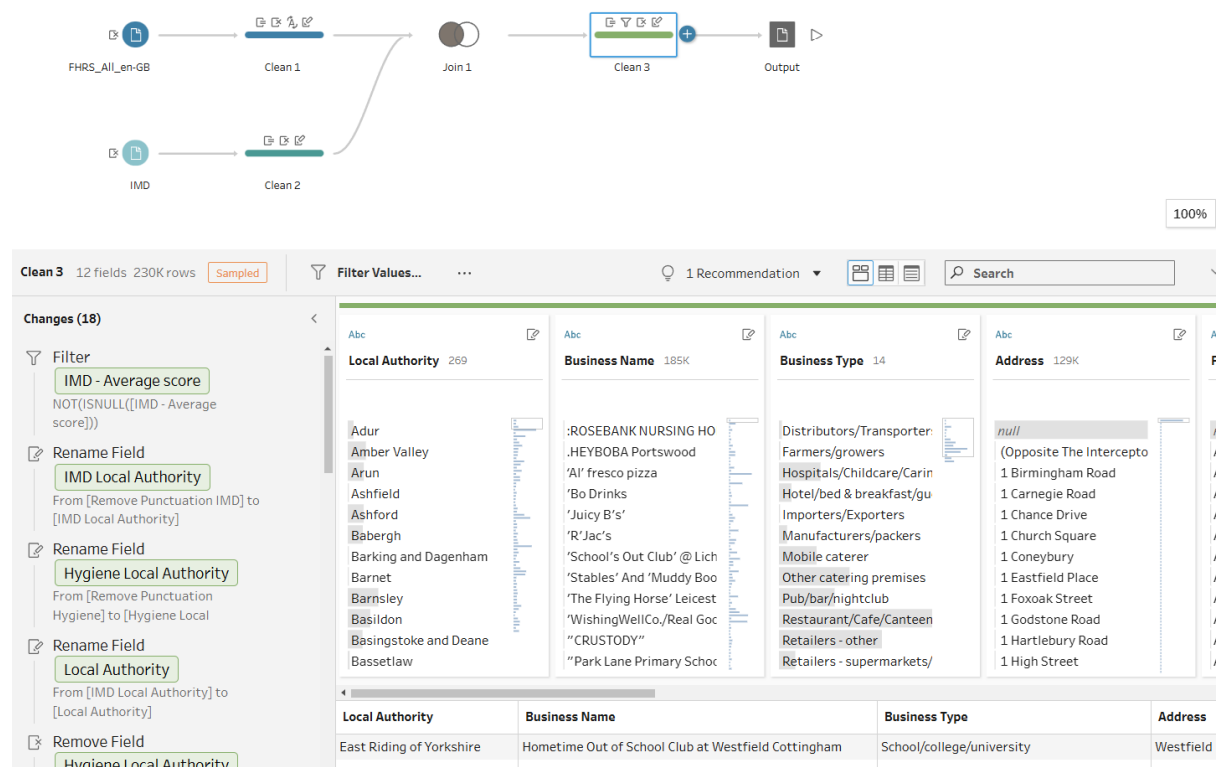
The two datasets were then joined using a left join, with the FHRs dataset as the main dataset. I chose a left join because the food hygiene records were the main focus of the project and I wanted to keep as many business records as possible while attaching on the deprivation dataset with the match local authority names.

Figure 5 Joining both datasets



After the join the next step of cleaning was to filter out records where there were null values in the deprivation score, hygiene rating and structural score as the fields containing this data provided no useful information in the final dataset. Then the remaining fields left were renamed into clearer and more understood naming convention, RatingValue becoming Hygiene Rating, Hygiene become Hygiene Score, Structural becoming Structural Score, and IMD – Average Score becoming Deprivation score. This made the final CSV easier to read and understand. After all of this processing was done, the final output consisted of 12 fields, over 400K rows and was then used as the data source for Tableau Cloud.

Figure 6 Combined dataset clean (Clean 3)

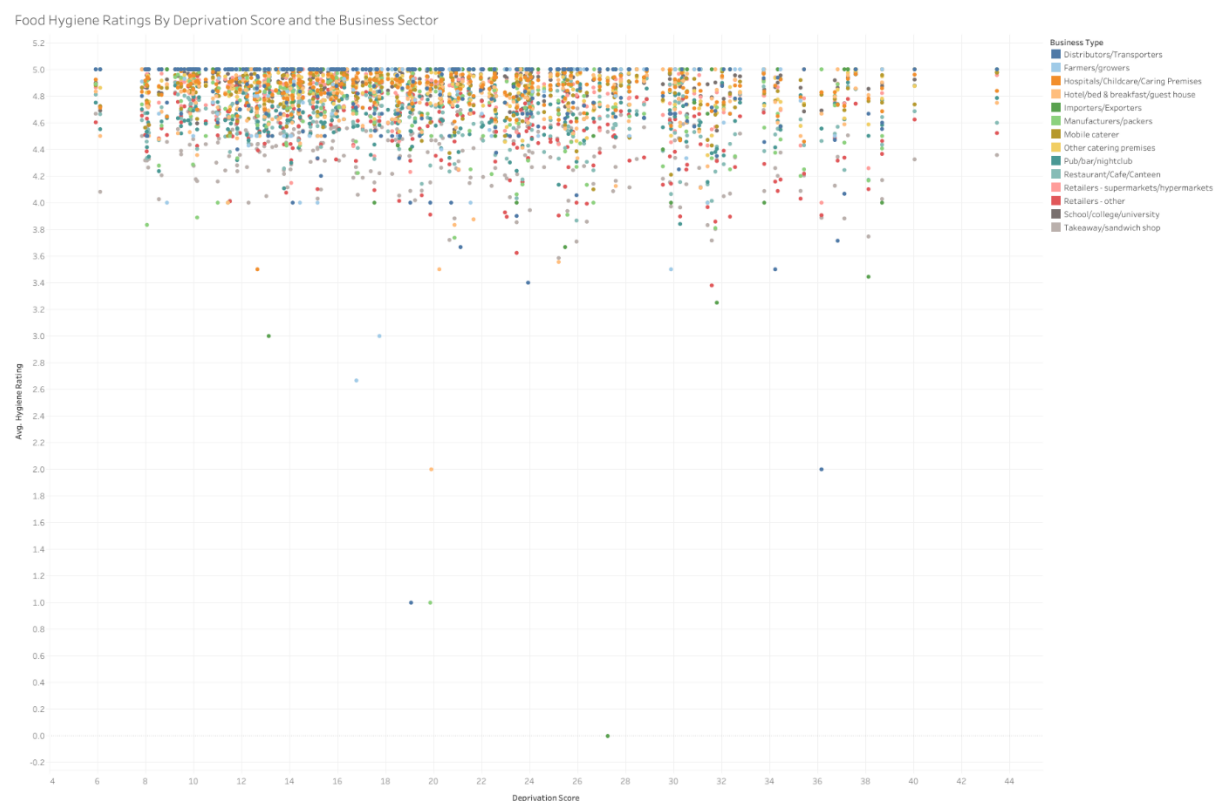


Visualisation

All visualisation was created using the Tableau Cloud which had the output data source from the automated process in Tableau Prep Builder. I used Tableau Cloud because it supported interactive dashboards, filters, maps, and charts without needing to build a full custom web application, which also fit well with Tableau Prep because the process output can be completely loaded into Tableau Cloud for analysis, keeping the entire process within the same single ecosystem.

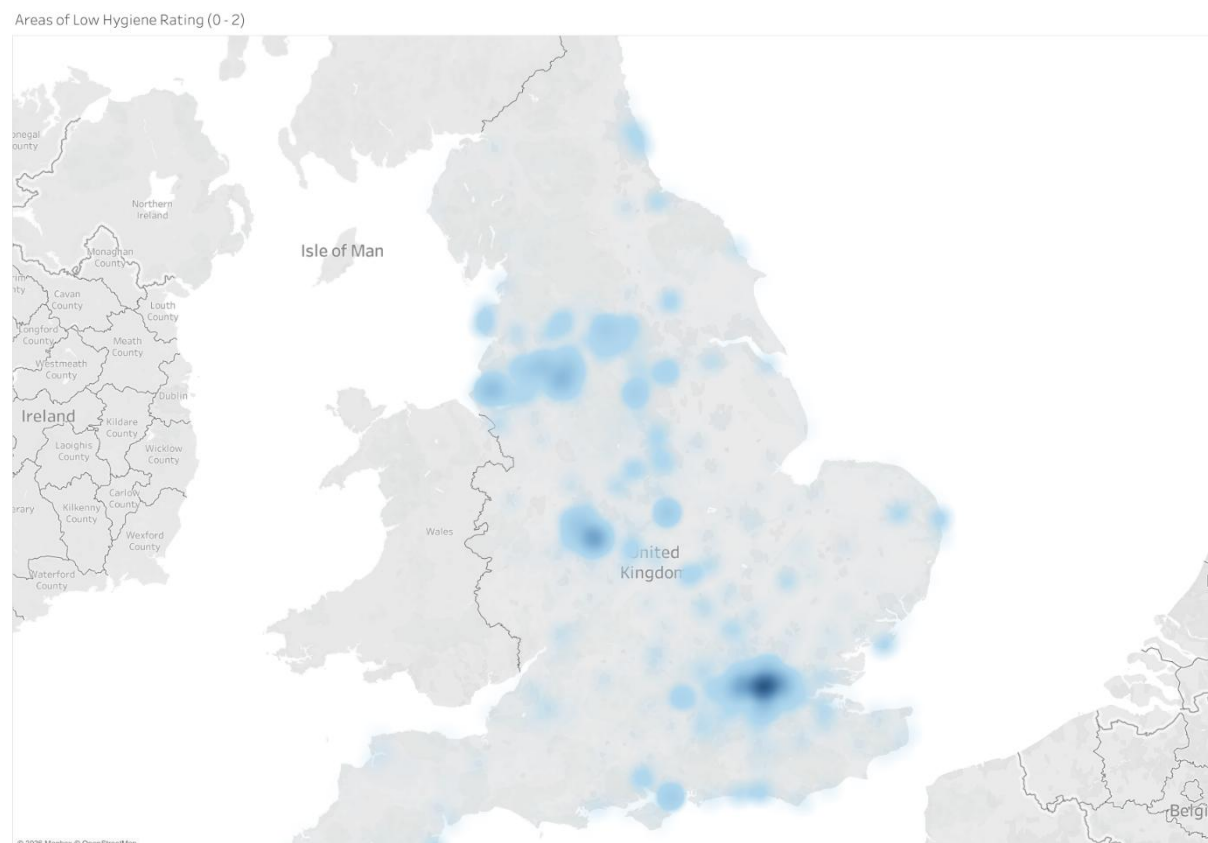
The first visualisation is a scatter plot showing food hygiene ratings by deprivation score and business type. The x-axis was the deprivation for each local authority, and the y-axis showed the average hygiene rating with each point coloured by the business type, which made it easier to compare each sector, such as takeaways, restaurants, schools etc. So, using the scatter plot helped show whether hygiene ratings change as deprivation scores increase and the colour provided some insight to each sector because it shows whether the pattern is across all food businesses are the same or if certain sectors perform differently.

Figure 7 Food Hygiene Ratings By Deprivation Score and the Business Sector plot



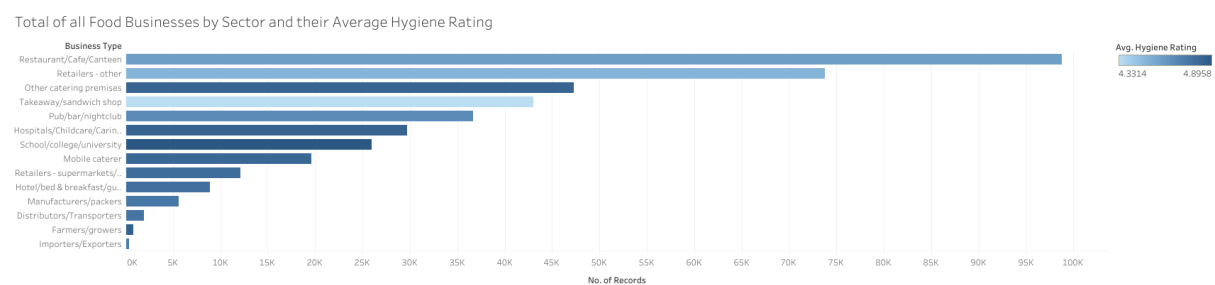
The second visualisation used a density map with business that had low hygiene ratings between 0 and 2. I chose the map because the FHRS dataset contains the latitudes and longitudes for businesses which made it possible to explore hygiene ratings across England. This is useful because low hygiene rating be easier to understand as clusters on the map rather than a large table and the map allows us to look at areas where low rated business appear more concentrated.

Figure 8 Areas of Low Hygiene Rating (0 - 2) map



The third visualisation is a bar chart showing the total number of food business by sector and coloured by the average hygiene rating which allows us and anyone else to understand the structure of the dataset. For example, restaurants, cafes and canteens make up the largest sector, while importers/exporters and farmers/growers have a lot less records within the dataset. The colouration of the chart added extra meaning to show how average hygiene rating differs between sectors. So, in the final dataset, takeaway/sandwich shops had one of the lowest average hygiene ratings, while schools, colleges and universities had one of the highest.

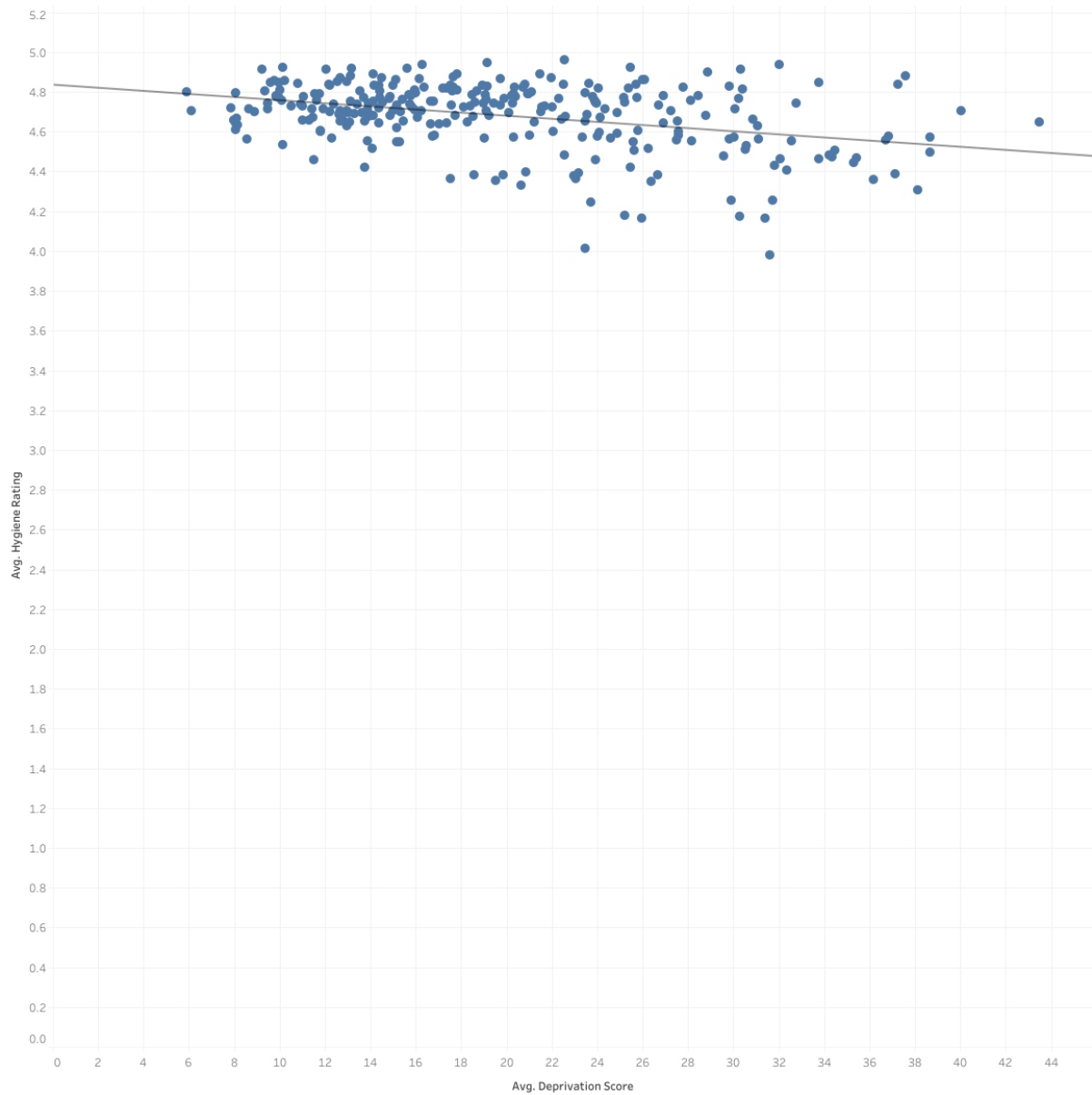
Figure 9 Total of all Food Businesses by Sector and their Average Hygiene Rating graph



The fourth visualisation shows the relationship between the deprivation score and the average food hygiene rating at each local authority level. Each point in the scatter plot represents the local authority, with the x-axis showing the average deprivation score and the y-axis showing the average hygiene rating, so I used these average values because each local authority does contain many food businesses which made the comparison between both easier. The trend line was also used, and it shows that it slopes slightly downwards which suggests that local authorities with higher deprivation scores often have slightly lower average hygiene ratings. Even looking at where the points are spread, it does show that deprivation is not the only factor affecting hygiene but does show a possible relationship with each other.

Figure 10 Correlation Between Deprived Areas and Food Hygiene plot

Correlation Between Deprived Areas and Food Hygiene



The dashboard includes interactivity through filtering such as business type and deprivation score which allows us to narrow the data down and explore different questions in this topic. For example, we can filter only takeaway/sandwich shops then analyse whether this sector shows a stronger pattern across deprivation scores.

A key strength in the visualisation used compared to non-interactive static charts is that by using a map, scatter plots, and comparing the sectors gives us multiple ways to understand the data and use different analysis with the data. A limitation is that some businesses could not be shown on the map because their latitude and longitude values were missing and deprivation is only measured at the local authority level, which leaves out specific areas and streets that may have higher deprivation and different hygiene scores, but being more generalised over the entire area.

Figure 11 Dashboard

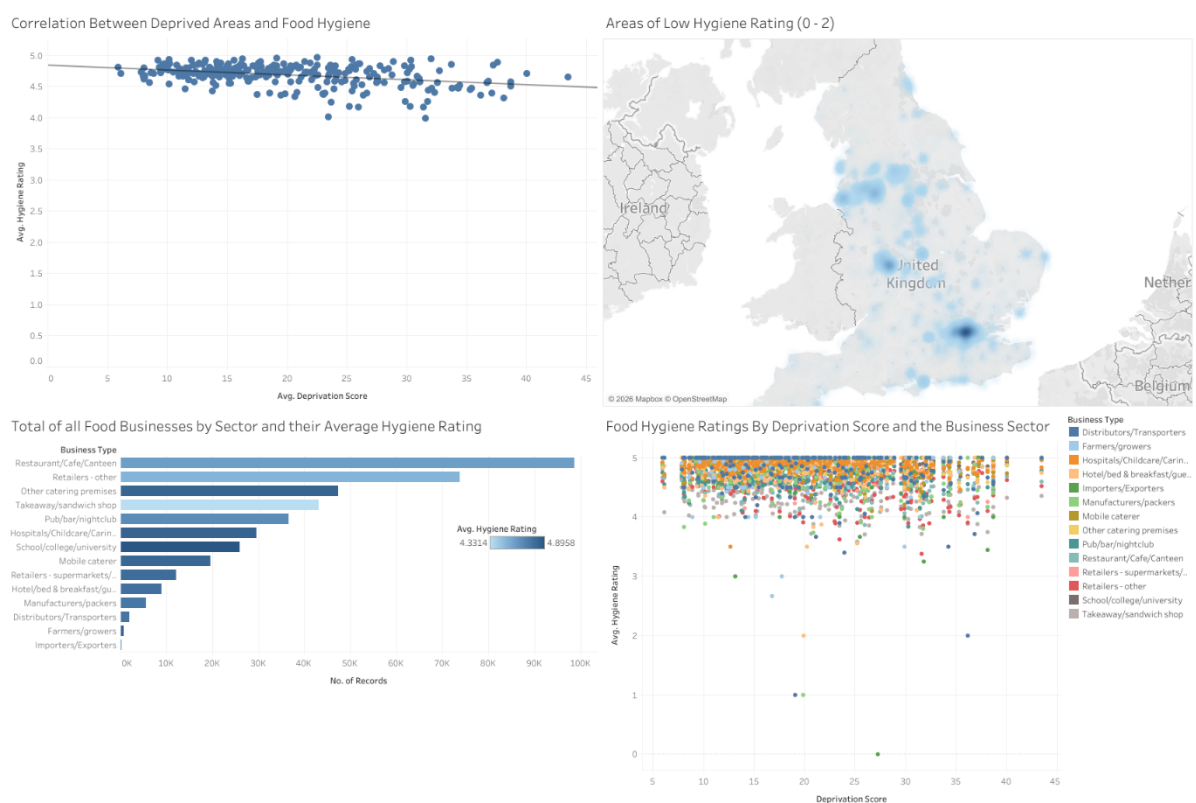
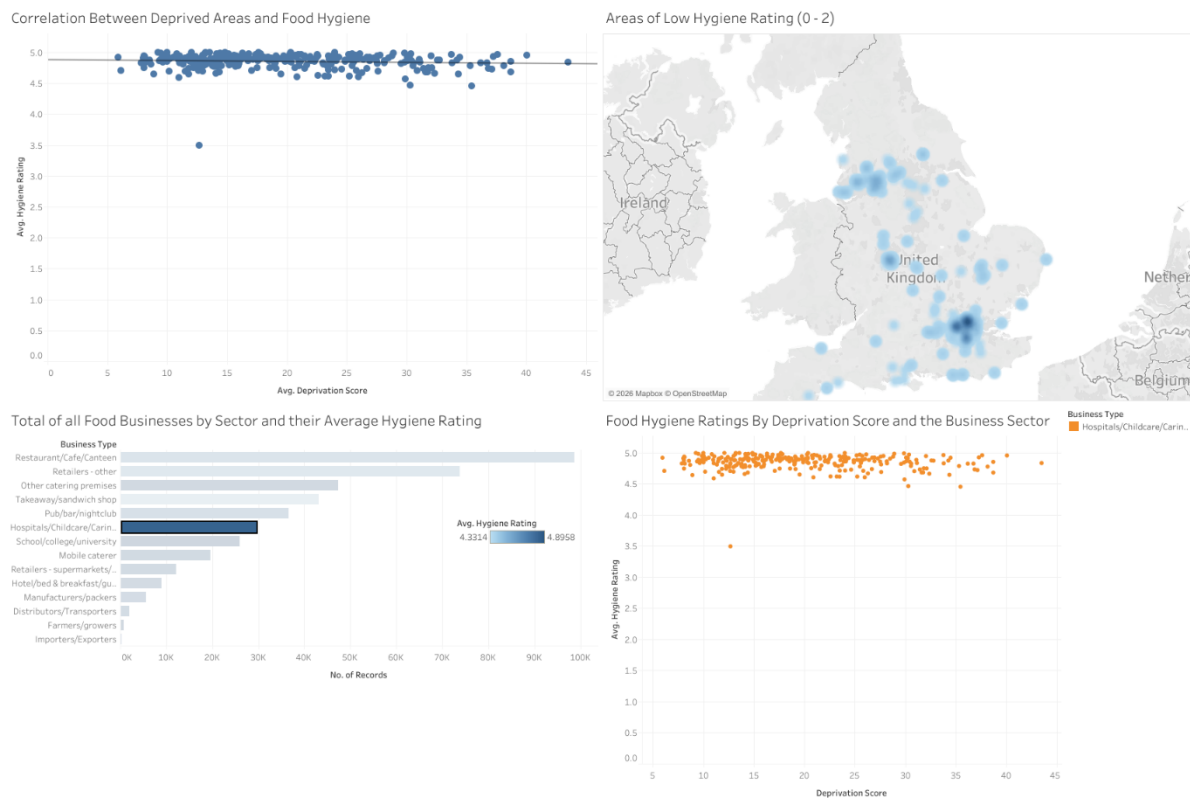


Figure 12 Interactive filter by Hospitals/Childcare/Caring Premises



I would say overall the visualisation allows us to explore the relationship between deprivation and food hygiene standards and shows that there appears to be a slight negative relationship between the deprivation score and average hygiene rating, while also highlighting any differences between business sectors and mapping clusters of low hygiene rated businesses.

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